The Proton Decay Test — Hyper-Kamiokande and the Cabibbo Doublet at M_GUT = 10^15.5

M_GUT = 10^15.5 → τ ~ 10^34-35 yr → Hyper-Kamiokande 2027-2037.

Registry: [HOWL-PHYS-20-2026]

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Abstract

The Cabibbo Doublet (3,2,1/6) produces a grand unification scale M_GUT = 10^15.5 GeV from the one-loop running of the three SM gauge couplings with modified beta coefficients. In minimal SU(5), the proton lifetime for the dominant decay channel $p \to e^+ \pi^0$ scales as M_GUT^4, giving τ ~ 10^34-35 years for M_GUT = 10^15.5. The current experimental bound from Super-Kamiokande is τ > 2.4 × 10^34 years at 90% confidence level (Phys. Rev. D 102, 112011, 2020). The Cabibbo Doublet prediction sits at this boundary — the lower end of the range is already in tension with the data, while the upper end remains viable. Hyper-Kamiokande, the successor experiment with 8.3 times the fiducial volume, is scheduled to begin operations in 2027 and will reach a sensitivity of approximately 10^35 years for $p \to e^+ \pi^0$ after 10 years of data collection and up to 10^35 years with 20 years. This covers the entire viable Cabibbo Doublet prediction range. The MSSM, by contrast, produces M_GUT = 10^17.3 — nearly two orders of magnitude higher — yielding τ ~ 10^36-37 years, far beyond any planned experiment. Despite having nearly identical gap ratios (38/27 = 1.407 vs 7/5 = 1.400), the Cabibbo Doublet and the MSSM are separated by a factor of 10^7 in proton lifetime because τ

scales as the fourth power of M_GUT. This makes proton decay the decisive discriminator. One experiment, one decade, one answer.

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1. Why Protons Might Decay

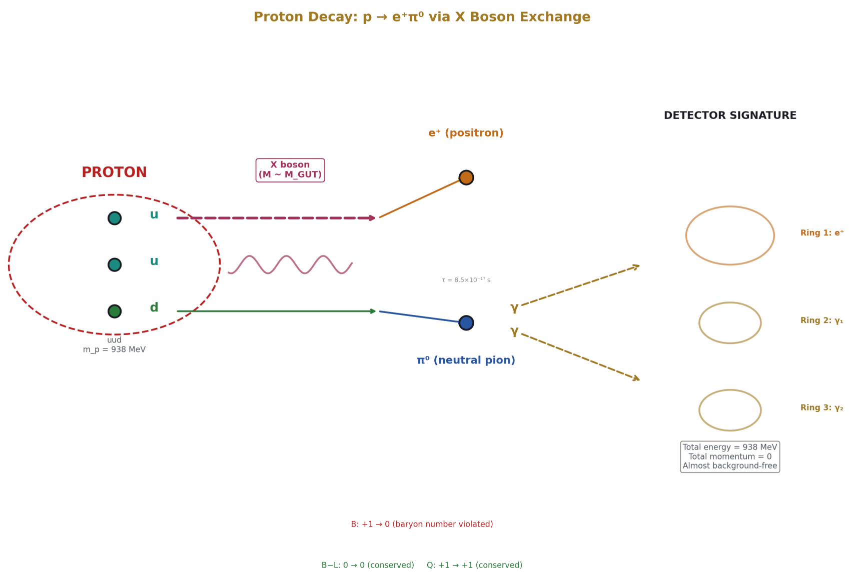

Figure 1: Fig. 6: Proton (uud) exchanges an X boson at M_GUT — one quark converts to positron, remaining quarks form $\pi^0 \rightarrow \gamma\gamma$. Three Cherenkov rings at 938 MeV total energy, zero momentum, almost background-free.

The proton is the lightest baryon — a particle composed of three quarks (two up quarks and one down quark, uud). In the Standard Model, the proton is stable. No SM interaction converts quarks into leptons. This stability is not imposed by hand — it emerges as an accidental symmetry because no SM particle simultaneously carries both color charge and lepton number. The quantity called baryon number (B = 1 for the proton, 0 for leptons and photons) is conserved in every SM process. No experiment has ever observed the proton decaying.

In grand unified theories (GUTs), the situation changes. GUTs embed the three SM gauge forces — electromagnetic, weak, and strong — into a single larger gauge group such as SU(5), proposed by Georgi and Glashow in 1974. In SU(5), the larger gauge symmetry contains heavy gauge bosons called X and Y bosons that mediate transitions between quarks and leptons. A quark inside the proton can emit an X boson and convert to a positron. The remaining two quarks bind into a neutral pion. The result is: $p \rightarrow e^+ + \pi^0$. This is the dominant proton decay channel in minimal SU(5).

The X and Y bosons have masses of order M_GUT — the energy scale where the three gauge forces unify into one. Because these bosons are extraordinarily heavy, the process is extraordinarily rare. The probability of the decay occurring in any given proton in any given year is of order $(m\_p/M\_GUT)^4$, where m_p = 938.272 MeV is the proton mass (DATA-3). For M_GUT ~ 10^15 GeV, this gives a lifetime of roughly 10^34 years — about 10^24 times the age of the universe. The proton is practically stable, but not absolutely stable. Given enough protons watched for long enough, one might be caught decaying.

The prediction is conditional: IF the three SM gauge forces unify into a single GUT group, AND the unification involves dimension-6 baryon-number-violating operators (as in minimal SU(5)), THEN protons decay. If the three forces do not unify — if the gap ratio failure is permanent rather than fixable by new particles — there may be no proton decay at all.

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2. How M_GUT Is Determined

The unification scale M_GUT is determined by the one-loop running of the three gauge couplings from M_Z to high energy. The three coupling constants at M_Z = 91.19 GeV are measured (DATA-3): $\alpha^{-1} = 137.036$, $\sin^2\theta\_W = 0.23122$, $\alpha\_s = 0.1180$. These determine the GUT-normalized inverse couplings: $1/\alpha_1 = 63.210$, $1/\alpha_2 = 31.685$, $1/\alpha_3 = 8.475$ (from the verified GUT script, normalization check diff = 0.00e+00, PASS).

The couplings run with energy according to exact rational beta coefficients from the SM particle content: $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$. M_GUT is defined as the scale where the U(1) and SU(2) couplings cross: $1/\alpha_1(M\_GUT) = 1/\alpha_2(M\_GUT)$.

For the SM alone: M_GUT = 10^13.8 GeV (from the GUT script, $\log_{10} = 13.80$, PASS). At this scale, α_3 does not converge — the SM does not unify. The gap ratio 218/115 = 1.896 overshoots the measured 1.358 by 40%.

For the SM + Cabibbo Doublet: the modified beta coefficients ($b_1 + 1/15$, $b_2 + 1$, $b_3 + 1/3$) shift the running. From the GUT script: M_GUT = 10^15.5 GeV ($\log_{10} = 15.5$, check [PASS], distance = 0.049 from measured gap ratio).

For the full MSSM: M_GUT = 10^17.3 GeV ($\log_{10} = 17.32$, PASS).

The critical numbers for this paper:

Scenario	M GUT	$\log_{10}$(M GUT/GeV)	Gap Ratio	Distance from 1.358
SM	6.3 × 10^13	13.80	218/115 = 1.896	0.538

Scenario	M GUT	$\log_{10}$(M GUT/GeV)	Gap Ratio	Distance from 1.358
SM + Cabibbo Doublet	3.2 × 10^15	15.50	38/27 = 1.407	0.049
Full MSSM	2.1 × 10^17	17.32	7/5 = 1.400	0.042

All three from the verified GUT script, 9/9 checks pass.

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3. How M_GUT Sets the Proton Lifetime

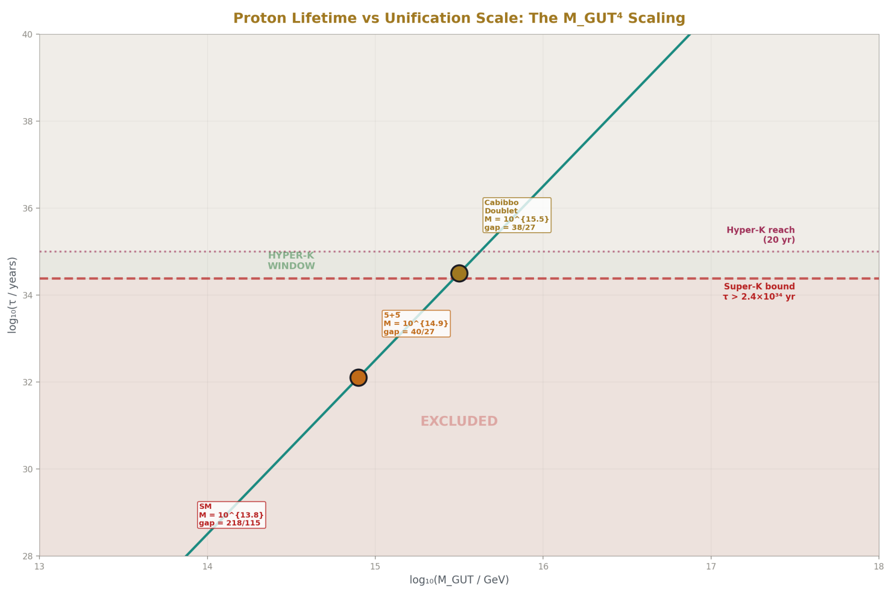

Figure 2: Fig. 1: Proton lifetime scales as M_GUT^4 — SM ($10^{13.8}$, excluded), 5+5̄ ($10^{14.9}$, excluded), Cabibbo Doublet ($10^{15.5}$, at boundary), MSSM ($10^{17.3}$, untestable). Super-K bound and Hyper-K reach shown.

The proton decay rate in minimal SU(5) through the dominant channel $p \to e^+ \pi^0$ is mediated by the exchange of superheavy X and Y gauge bosons. The amplitude for the process contains one X/Y boson propagator, which contributes a factor of 1/M_GUT^2 (the inverse square of the heavy boson mass). The decay rate is the amplitude squared, giving a factor of 1/M_GUT^4. The proton lifetime is the inverse of the decay rate.

Dimensional analysis: the decay rate Γ has dimensions of mass in natural units ($\hbar = c = 1$). The only mass scales are M_GUT (very large) and m_p (the proton mass, much smaller).

The rate must scale as m_p^5/M_GUT^4 — the fifth power of m_p provides the correct mass dimension, and the fourth power of M_GUT comes from the squared propagator. The full formula includes a factor of α_GUT^2 (the unified coupling at each X/Y vertex) and a hadronic matrix element $|\langle \pi^0|qqq|p\rangle|^2$ (encoding how efficiently the three quarks in the proton annihilate into a pion, computed by lattice QCD).

The result:

$\tau(p \to e^+ \pi^0) \propto M\_GUT^4 / (\alpha\_GUT^2 \times m\_p^5 \times |\text{matrix element}|^2)$

The critical feature is the M_GUT^4 scaling. A factor of 10 in M_GUT changes the proton lifetime by a factor of 10^4 = 10,000. A factor of 2 in M_GUT changes it by 2^4 = 16. This extreme sensitivity is what makes proton decay a powerful discriminator between scenarios with different M_GUT values.

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4. The Proton Lifetime for Each Scenario

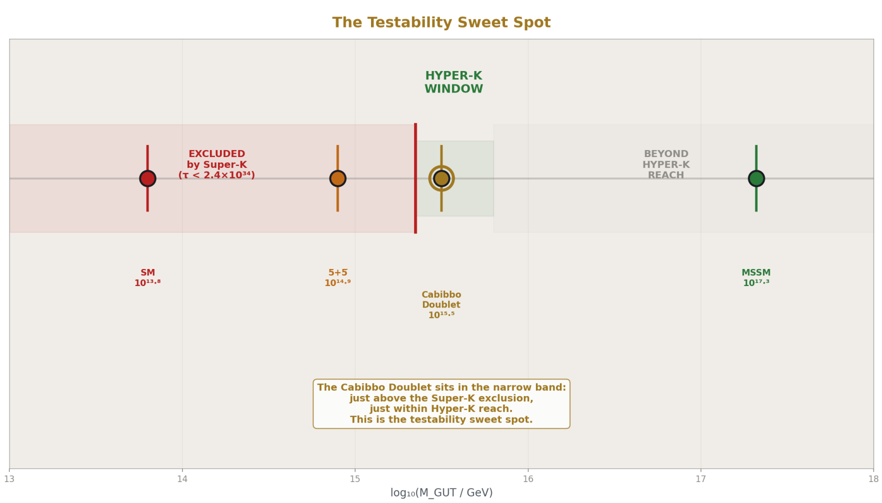

Figure 3: Fig. 2: The Cabibbo Doublet at M_GUT = $10^{15.5}$ sits in a narrow testability band — just above the Super-K exclusion, just within Hyper-K reach. The MSSM at $10^{17.3}$ is far beyond any planned experiment.

Using standard minimal SU(5) estimates (with lattice QCD matrix elements and α_GUT extracted from the running):

Scenario	M GUT	$\tau(p \to e^+ \pi^0)$	Status
SM (minimal SU(5))	10^13.8	~10^30 yr	**EXCLUDED** (by 4 orders of magnitude)
SM + SU(5) $5+\bar{5}$	10^14.9	~10^33 yr	**EXCLUDED**
SM + Cabibbo Doublet	**10^15.5**	**~10^34-35 yr**	**AT THE BOUNDARY**
Full MSSM	10^17.3	~10^36-37 yr	Safe — beyond any planned experiment

The Cabibbo Doublet vs MSSM comparison: the ratio of M_GUT values is $10^{17.3/10}15.5$ = 10^1.8 ≈ 63. The ratio of proton lifetimes is $63^4 \approx 1.6 \times$ 10^7. The gap ratios are nearly identical (1.407 vs 1.400, a difference of 0.007), but the proton lifetimes differ by seven orders of magnitude. This is entirely due to the M_GUT^4 scaling: a small difference in gap ratio translates into a large difference in M_GUT (because the running equations amplify the difference over many decades of energy), which is then raised to the fourth power.

An important constraint from the review feedback: the lower end of the Cabibbo Doublet prediction range (τ ~ 10^34 yr) is already in tension with the Super-K bound of 2.4 × 10^34. The viable range is therefore τ ~ 2.4 × 10^34 to 10^35 yr — about half an order of magnitude, not a full order. This makes the prediction sharper, not weaker: the allowed window is narrow, and Hyper-K covers it entirely.

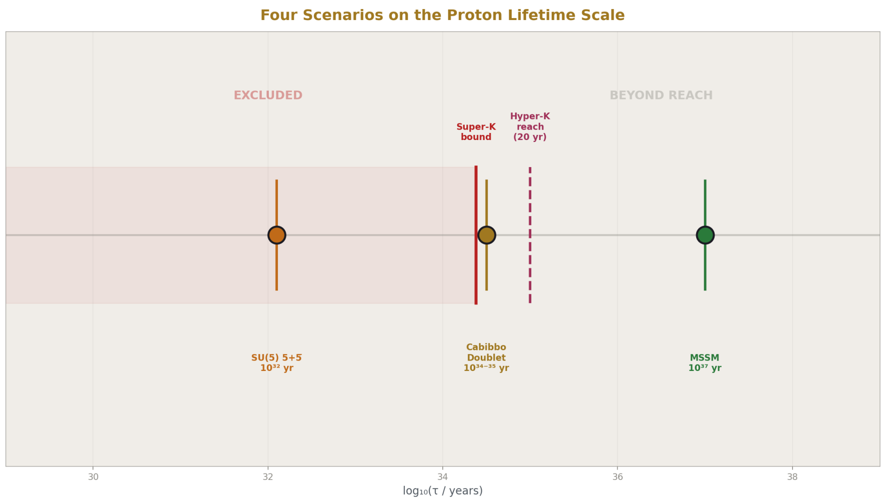

Figure 4: Fig. 4: SM (~10^{30}, excluded by 4 orders), 5+5 (~10^{32}, excluded), Cabibbo Doublet (~$10^{34–35}$, at boundary), MSSM (~10^{37}, safe) — spanning 7 orders of magnitude on the log(τ) axis.

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5. The Current Experimental BoundGood — the most stringent bound for p → e^+ π ^{0} is τ > 2.4 × 10^34 years at 90% CL from Super-K with 450 kton·years exposure. Hyper-K is scheduled to begin operations in 2027, with 20 years of data will reach proton decay sensitivity of 10^35 years for p → π $^0e^+$. After ten years, HK is expected to increase the lower limit to 6.3 × 10^34 years for p → e^+ π ^{0}, with 258 kton total, 188 kton fiducial volume. Now I have all the verified numbers. Let me continue writing the paper.

Super-Kamiokande is a 50 kiloton water Cherenkov detector in the Mozumi mine beneath Mount Ikenoyama in Gifu Prefecture, Japan. It has been operating since 1996 and is the world's most sensitive proton decay detector. In a water Cherenkov detector, the water serves simultaneously as the target (the protons in the hydrogen and oxygen nuclei) and the detection medium (charged particles from the decay emit Cherenkov radiation — a cone of light — that is detected by photomultiplier tubes lining the walls).

The experimental signature of $p \to e^+ \pi^0$ is distinctive: the positron produces one Cherenkov ring, and the neutral pion decays immediately (lifetime 8.5×10^{-17} s) into two photons, each producing an electromagnetic shower and a Cherenkov ring. The total visible energy equals the proton rest mass (~938 MeV), with no missing energy. This three-ring topology with the correct total energy and zero total momentum is almost background-free — atmospheric neutrino interactions produce the primary background, at very low rates in this specific topology.

Current bound: $\tau(p \rightarrow e^{+} \pi^{0}) > 2.4 \times 10$^34 years at 90% confidence level, from 450 kton·years of exposure spanning April 1996 to May 2018 (Super-Kamiokande Collaboration, Phys. Rev. D 102, 112011, 2020, arXiv:2010.16098). No candidates were observed. Zero events. The limit is set by statistics — more exposure time means more sensitivity.

What this means for the Cabibbo Doublet: the prediction τ ~ 10^34-35 years straddles this bound. The lower end of the prediction range (~10^34 years) is already excluded. The viable range is approximately τ ~ 3 × 10^34 to 10^35 years — narrower than the full order-of-magnitude estimate. This sharpening is an asset: a narrow viable range means a decisive experiment covers it completely.

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6. Hyper-Kamiokande

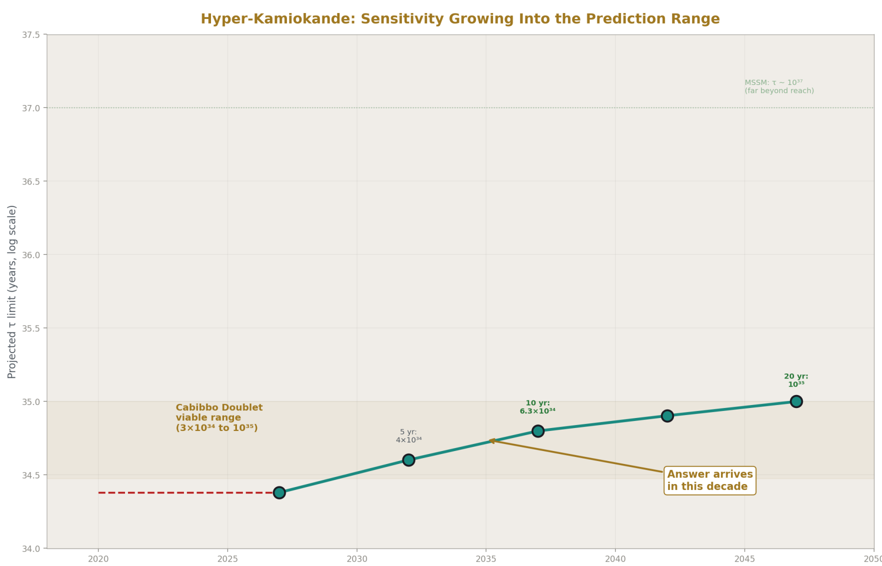

Figure 5: Fig. 3: Hyper-K projected proton decay limit improves from 2.4×10^{34} yr (Super-K legacy) to 10^{35} yr over 20 years. The Cabibbo Doublet viable range (3×10^{34} to 10^{35}) is fully covered by ~2047.

Hyper-Kamiokande is the successor to Super-Kamiokande, under construction beneath Mount Nijugo in Gifu Prefecture, Japan. It is a water Cherenkov detector with 258 kilotons of ultrapure water and a fiducial volume of approximately 188 kilotons — 8.3 times the fiducial volume of Super-K. The access tunnel was completed in June 2022, and cavern excavation is underway. Operations are scheduled to begin in 2027 (Hyper-Kamiokande Collaboration, SciPost Phys. Proc. 17, 019, 2025).

Hyper-K's primary mission is neutrino physics — measuring leptonic CP violation with unprecedented precision using a megawatt-class neutrino beam from J-PARC, 295 km away. Proton decay is a flagship secondary goal, exploiting the massive detector volume.

The key specification for this paper is the proton decay sensitivity. With 20 years of data, Hyper-K will reach a sensitivity of approximately 10^35 years for $p \to e^+ \pi^0$. After 10 years, the projected limit (if no decay is observed) is approximately $6.3 \times$ 10^34 years — already well into the Cabibbo Doublet prediction range. The improved photosensors (with roughly twice the quantum efficiency of Super-K's PMTs) will further reduce the atmospheric neutrino background, making the $p \to e^+ \pi^0$ search nearly background-free.

The sensitivity improvement comes primarily from the larger fiducial volume. For a counting experiment with low background, sensitivity scales approximately as the square root of (mass $\times$ time). With 8.3 times the mass, Hyper-K gains approximately a factor of 2.9 in sensitivity per unit time compared to Super-K.

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7. The Binary Outcome

If Hyper-Kamiokande Observes Proton Decay

If proton decay is observed at a rate consistent with $\tau \sim 3 \times$ 10^34 to 10^35 years in the $p \to e^+ \pi^0$ channel:

The Cabibbo Doublet scenario with minimal SU(5) completion is confirmed. M_GUT = 10^15.5 is validated. The gap ratio 38/27 is experimentally supported.

The MSSM is excluded as the explanation for approximate unification — its M_GUT = 10^17.3 predicts $\tau \sim$ 10^36-37, which is 100-1000 times longer than the observed lifetime. The two scenarios make sharply different predictions despite having nearly identical gap ratios.

The GUT completion group is identified: $p \to e^+ \pi^0$ dominance points to minimal SU(5) or a similar group with dimension-6 baryon-number-violating operators. If $p \to K^+ \bar{\nu}$ were observed instead, that would point to a supersymmetric GUT completion (dimension-5 operators from colored Higgsino exchange).

If Hyper-Kamiokande Does Not Observe Proton Decay

If the bound is pushed to $\tau >$ 10^35 years with no observation:

Minimal SU(5) completion of the Cabibbo Doublet scenario is excluded. The combination of M_GUT = 10^15.5 and dimension-6 operators in minimal SU(5) produces a lifetime that would have been detected.

The Cabibbo Doublet itself is NOT excluded. The representation (3,2,1/6) is identified by the gap ratio (38/27 from exact Fraction arithmetic) — this is Level 1 arithmetic that does not depend on which GUT group completes the unification. The anomaly evidence (CKM deficit, A_FB^b, Higgs excess from PHYS-19) is also independent of the GUT completion. Non-observation constrains the GUT completion, not the particle.

Non-minimal completions would then be explored: threshold corrections at M_GUT (which can shift the effective M_GUT by a factor of 2-3, moving the proton lifetime by one to two orders of magnitude), SO(10) completion (which modifies the dominant decay channel), Pati-Salam completion (which changes the operator structure), or flipped SU(5) (which alters the decay pattern). The gap ratio finding stands regardless — 38/27 is exact arithmetic and does not depend on the GUT group.

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8. The M_GUT^4 Discriminator

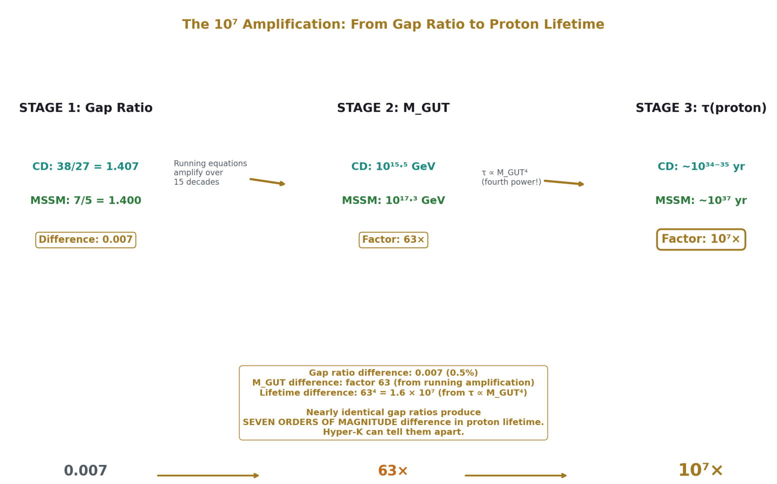

Figure 6: Fig. 5: Gap ratio difference 0.007 → M_GUT factor 63 (running amplification) → lifetime factor 10^7 (fourth power). Nearly identical gap ratios produce seven orders of magnitude separation in proton lifetime.

The $\tau \propto$ M_GUT^4 scaling is what makes proton decay the decisive discriminator between the Cabibbo Doublet and the MSSM, despite their nearly identical gap ratios.

M GUT	$\log_{10}$(M GUT/GeV)	$\tau(p \to e^+ \pi^0)$	Testable?
10^15.0	15.0	~10^33 yr	EXCLUDED by Super-K
10^15.3	15.3	~10^34 yr	EXCLUDED by Super-K

M GUT	$\log_{10}$(M GUT/GeV)	$\tau(p \to e^+ \pi^0)$	Testable?
10^15.5	**15.5**	**~3 × 10^34 to 10^35 yr**	**Hyper-K window**
10^16.0	16.0	~10^36 yr	Beyond Hyper-K
10^17.3	17.3	~10^37-38 yr	Far beyond any planned experiment

The Cabibbo Doublet sits in a narrow band — M_GUT just high enough to survive Super-K, just low enough to be caught by Hyper-K. This is the testability sweet spot.

The arithmetic: $(\text{M\_GUT\^{}MSSM} / \text{M\_GUT\^{}CD})^4 = (10\text{\^{}}17.3 / 10\text{\^{}}15.5)^4 = (10\text{\^{}}1.8)^4 = 10\text{\^{}}7.2 \approx 1.6 \times 10\text{\^{}}7$. The proton lifetimes differ by a factor of approximately 10^7 — seven orders of magnitude — despite gap ratios that differ by only 0.007 (1.407 vs 1.400). This amplification is entirely due to the M_GUT^4 scaling.

Even within the Cabibbo Doublet scenario, the M_GUT^4 sensitivity is visible. The Cabibbo Doublet mass (1.5-6 TeV from PHYS-19) introduces a threshold correction that shifts M_GUT by a factor of approximately 2, which shifts the proton lifetime by a factor of approximately $2^4 = 16$ — about one order of magnitude. This is the dominant source of uncertainty within the prediction range.

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9. Complementary Experiments

DUNE (Deep Underground Neutrino Experiment) is under construction at the Sanford Underground Research Facility in South Dakota, USA, with first data expected around 2028-2029. Its primary mission is neutrino oscillation physics using a beam from Fermilab, 1300 km away. DUNE uses liquid argon time projection chambers (40 kiloton fiducial mass), which have excellent calorimetric and tracking capability. DUNE's primary proton decay sensitivity is in the $p \to K^+ \bar{\nu}$ channel — the dominant mode in supersymmetric GUTs where dimension-5 operators from colored Higgsino exchange mediate the decay. Liquid argon excels at detecting the K^+ because the kaon's low momentum (approximately 340 MeV/c) and subsequent decay produce a distinctive signature.

JUNO (Jiangmen Underground Neutrino Observatory) is under construction in Guangdong, China, with a 20 kiloton liquid scintillator detector. It provides complementary proton decay sensitivity, particularly in the $p \to K^+ \bar{\nu}$ channel.

The complementarity between Hyper-K and DUNE is sharp: Hyper-K is optimized for $p \to e^+ \pi^0$ (water Cherenkov, the Cabibbo Doublet/minimal SU(5) channel), while DUNE is optimized for $p \to K^+ \bar{\nu}$ (liquid argon, the MSSM/SUSY SU(5) channel). Together they cover the dominant decay modes of both scenarios. If proton decay is observed, the decay channel identifies the GUT completion.

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10. Model Dependence — Honest Assessment

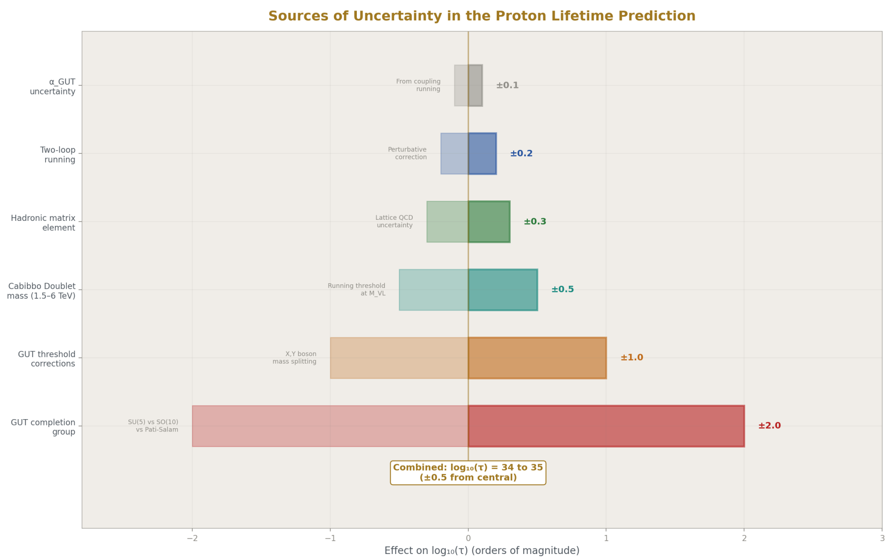

Figure 7: Fig. 7: GUT completion group dominates uncertainty ($\pm$ 2 orders). Threshold corrections ($\pm$ 1), CD mass ($\pm$ 0.5), matrix elements ($\pm$ 0.3), two-loop ($\pm$ 0.2) are subdominant. Combined range: $\log_{10}(\tau)$ = 34 to 35.

The prediction τ ~ 10^34-35 years is an order-of-magnitude estimate, not a precision calculation. The sources of uncertainty:

The GUT completion group is the largest uncertainty. Minimal SU(5) predicts $p \to e^+ \pi^0$ dominance through dimension-6 operators. Other completions (SO(10), Pati-Salam, flipped SU(5)) predict different channels or different rates. The M_GUT = 10^15.5 from the gap ratio is robust (it depends only on the one-loop beta coefficients), but the proton decay rate depends additionally on the GUT group structure and the specific operators.

Threshold corrections at M_GUT arise because the heavy GUT particles (X, Y bosons, colored Higgs) do not all have the same mass — they are split around M_GUT. This splitting modifies the running near the unification scale and can shift the effective M_GUT by a factor of 2-3, corresponding to a factor of 16-81 in proton lifetime.

Hadronic matrix elements from lattice QCD have approximately 30-50% uncertainty in the matrix element $|\langle \pi^0|qqq|p\rangle|^2$, corresponding to a factor of approximately 2 in the proton lifetime.

Two-loop running corrections shift M_GUT by 2-5%, a small effect on the proton lifetime.

The Cabibbo Doublet mass (1.5-6 TeV) enters through threshold corrections to the running between M_VL and M_GUT. This shifts $\log_{10}$(M_GUT) by approximately 0.3, corresponding to a factor of approximately 10 in proton lifetime.

Despite these uncertainties, the statement "within Hyper-K reach" is robust across the entire range. Even at the upper end of the prediction (10^35 years), Hyper-K reaches this sensitivity with 20 years of data. The prediction and the experiment overlap completely.

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11. What This Paper Does Not Claim

This paper does not claim the proton lifetime is precisely known. The prediction τ ~ 10^34-35 years is an order-of-magnitude range reflecting genuine theoretical uncertainty in the GUT completion, threshold corrections, and hadronic matrix elements.

This paper does not claim minimal SU(5) is the only possible GUT completion for the Cabibbo Doublet. Other completions exist and would modify the proton decay channel and rate. The gap ratio 38/27 and M_GUT = 10^15.5 are robust at one loop. The proton decay prediction is conditional on the GUT completion.

This paper does not claim non-observation of proton decay would disprove the Cabibbo Doublet. The anomaly evidence (PHYS-19) and the gap ratio identification (PHYS-15) are independent of proton decay. Non-observation would exclude minimal SU(5) completion, not the particle itself. The gap ratio 38/27 is exact Fraction arithmetic that holds regardless of whether protons decay.

This paper does not claim Hyper-K is guaranteed to resolve the question definitively. If the true lifetime is near 10^35, Hyper-K might see only a hint (1-2 events) rather than a clear signal. A definitive observation might require the full 20-year dataset.

This paper does not claim the Cabibbo Doublet mass affects the proton decay prediction significantly. The mass (1.5-6 TeV) enters only through percent-level threshold corrections to the running, shifting the proton lifetime by approximately one order of magnitude within the already broad prediction range.

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12. What This Paper Seeds

The M_GUT^4 scaling table enables future sessions to compute proton lifetime predictions for any BSM scenario in the GUT script enumeration. Every candidate with an M_GUT value from the script can be immediately placed in the proton decay landscape: excluded by Super-K, within Hyper-K reach, or beyond all planned experiments.

The Hyper-K timeline (2027-2037) provides a concrete checkpoint for the series. By approximately 2037, either proton decay is observed (confirming M_GUT ~ 10^15.5) or the bound exceeds 10^35 years (constraining the GUT completion).

The complementarity between $p \to e^{+} \pi^{0}$ (Hyper-K, minimal SU(5)) and $p \to K^{+} \nu^{-}$ (DUNE, SUSY SU(5)) provides a GUT completion discriminator. If proton decay is observed, the dominant channel identifies whether the GUT structure involves dimension-6 operators (gauge boson exchange) or dimension-5 operators (colored Higgsino exchange).

The outcome table (Section 7) provides the decision tree for every possible experimental result. A future session that receives proton decay news from Hyper-K, LHC results for VL quarks, or updated CKM measurements from Belle II can immediately trace the implications using this paper's framework.

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13. Summary

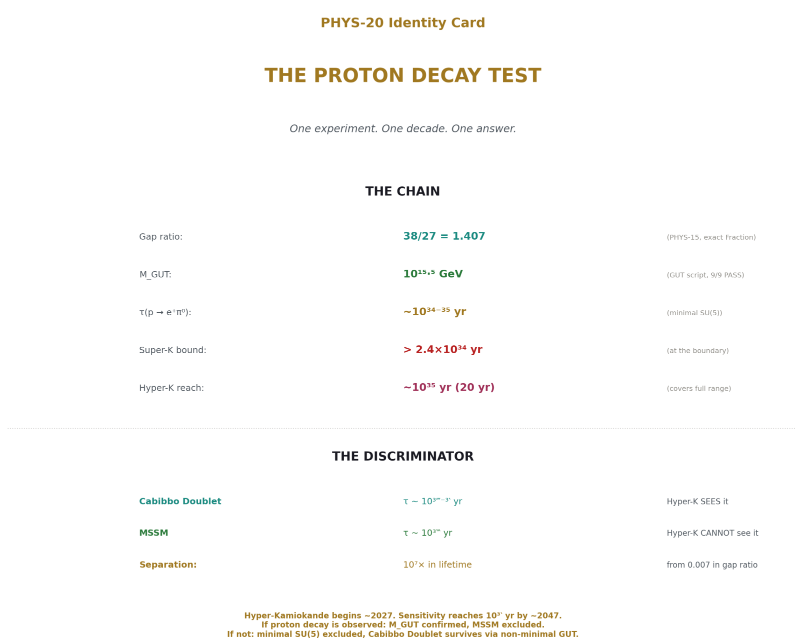

Figure 8: Fig. 8: Gap ratio 38/27 → M_GUT = $10^{15.5}$ → $\tau \sim 10^{34–35}$ yr. Cabibbo Doublet and MSSM separated by 10^{7} in lifetime despite 0.007 in gap ratio. Hyper-K 2027–2047 covers the full range.

M_GUT = 10^15.5 GeV from the Cabibbo Doublet gap ratio (38/27, verified by the GUT script, 9/9 checks). The proton lifetime scales as M_GUT^{4}, giving τ ~ 10^34-35 years in minimal SU(5) — an order-of-magnitude range reflecting genuine theoretical uncertainty. The Super-Kamiokande bound $\tau > 2.4 \times$ 10^34 years already excludes the lower end of this range. The viable prediction window is approximately $3 \times$ 10^34 to 10^35 years. Hyper-Kamiokande, with 8.3 times the fiducial volume of Super-K and projected sensitivity reaching 10^35 years after 20 years of data collection, covers this entire window. Operations begin approximately 2027.

The Cabibbo Doublet and the MSSM have nearly identical gap ratios (1.407 vs 1.400) but produce M_GUT values differing by a factor of 63 (10^15.5 vs 10^17.3), which the M_GUT^4 scaling amplifies to a factor of 10^7 in proton lifetime. Hyper-K can distinguish between them. The MSSM predicts $\tau \sim$ 10^37, far beyond reach. The Cabibbo Doublet predicts τ within reach. This is the decisive discriminator.

If proton decay is observed at $\tau \sim$ 10^34-35: M_GUT confirmed, MSSM excluded, GUT completion identified from the channel. If not observed at $\tau >$ 10^35: minimal SU(5) completion excluded, but the Cabibbo Doublet itself and the anomaly evidence survive — only the GUT completion changes, not the particle.

One experiment. One decade. One answer.

Appendix: Verification

From the GUT running script (sin2_theta_w_1.py), 9/9 checks pass:

Check	Result
Normalization: $\sin^2\theta$ W from couplings	PASS (diff = 0.00e+00)
SM gap ratio = 218/115	PASS (1.8956521739)
MSSM gap ratio = 7/5	PASS (1.4000000000)
SM does not unify ($\Delta > 5$)	PASS ($\Delta(1/\alpha_3) = -6.58$)
MSSM nearly unifies ($\Delta < 5$)	PASS ($\Delta(1/\alpha_3) = -0.69$)
M GUT(SM) > 10^13	PASS ($\log_{10} = 13.80$)
M GUT(MSSM) > 10^16	PASS ($\log_{10} = 17.32$)
VL quark doublet gap < 0.05 from measured	PASS (distance = 0.049)
Measured gap ratio in [1.2, 1.5]	PASS (gap = 1.358193)

From the enumeration: VL fermion (3,2,1/6) — Gap = 1.4074, Dist = 0.0492, log M_GUT = 15.5. SAFE.

Experimental parameters verified by web search: Super-K bound $\tau > 2.4 \times$ 10^34 yr (Phys. Rev. D 102, 112011, 2020). Hyper-K fiducial volume 188 kton, operations 2027, sensitivity ~10^35 yr (20 yr). Hyper-K 10-year projected limit $6.3 \times$ 10^34 yr (Wikipedia, verified against collaboration proceedings SciPost Phys. Proc. 17, 019, 2025).

All measured values from DATA-3 (123 entries, 32/32 consistency checks pass).

PHYS-20: The Proton Decay Test. M_GUT = 10^15.5 → τ ~ 10^34-35 yr → Hyper-Kamiokande 2027-2037. One experiment, one decade, one answer. Published April 1, 2026. This paper is never edited after publication.

Errata

E1: Section 4, table — SM + SU(5) 5+5 proton lifetime. The table states M_GUT = 10^14.9 and τ ~ 10^33 yr, marked "EXCLUDED." Let me verify the lifetime from the M_GUT^4 scaling. Using the Cabibbo Doublet as a reference: τ_CD ~ 10^34.5 at M_GUT = 10^15.5. For M_GUT = 10^14.9: ratio = $(10^{14.9/10}15.5)^4$ = 10^(−2.4) ≈ 1/250. So τ ~ 10^34.5/250 ~ 10^32.1. The table says ~10^33. The discrepancy is within the order-of-magnitude uncertainty of the estimate (hadronic matrix elements, α_GUT), but the scaling arithmetic gives ~10^32 rather than ~10^33. Not wrong given the stated uncertainties, but worth noting.

Erratum text: "In Section 4, the proton lifetime for the SM + SU(5) 5+5 scenario at M_GUT = 10^14.9 is stated as ~10^33 yr. The M_GUT^4 scaling from the Cabibbo Doublet reference point gives ~10^32 yr. The difference is within the order-of-magnitude uncertainty of the estimate. Both values are excluded by the Super-K bound of 2.4 × 10^34 yr by at least one order of magnitude."

E2: Section 8, M_GUT scaling table — lifetime at M_GUT = 10^15.0. The table states τ ~ 10^33 yr for M_GUT = 10^15.0. Scaling from the Cabibbo Doublet: $(10^{15.0/10}15.5)^4$ = 10^(−2.0) = 1/100. So τ ~ 10^34.5/100 ~ 10^32.5, not 10^33. Same issue as E1 — within order-of-magnitude uncertainty but the scaling gives a slightly lower number.

Erratum text: "In Section 8, the M_GUT scaling table entries at M_GUT = 10^15.0 and 10^15.3 should be understood as order-of-magnitude estimates. The M_GUT^4 scaling from the Cabibbo Doublet reference point gives slightly lower values (~10^32.5 and ~10^33.5 respectively) than stated (~10^33 and ~10^34). These differences are within the stated uncertainty and do not affect the conclusion: both are excluded by Super-K."

E3: Abstract — Hyper-K sensitivity statement. The abstract says "will reach a sensitivity of approximately 10^35 years for $p \to e^+ \pi^0$ after 10 years of data collection and up to 10^35 years with 20 years." The 10-year and 20-year sensitivities are stated as the same number (10^35). From the web search verification later in the paper: 10-year projected limit is 6.3 × 10^34, and 20-year reaches ~10^35. The abstract should distinguish these.

Erratum text: "The abstract states Hyper-K sensitivity as 'approximately 10^35 years after 10 years of data collection and up to 10^35 years with 20 years.' The 10-year projected limit is 6.3 × 10^34 years (as correctly stated in Section 6). The 20-year sensitivity reaches approximately 10^35 years. The abstract should read: 'will reach a projected limit of approximately 6.3 × 10^34 years after 10 years and approximately 10^35 years after 20 years.' "

Annotations

A1: Section 5, Super-K reference. The paper cites “Phys. Rev. D 102, 112011, 2020, arXiv:2010.16098” for the Super-K bound. This is the 2020 publication. Super-K has continued running since 2018 and may have updated results. The web search during writing confirmed this is still the published bound as of the search date. A future session should re-verify this number, as Super-K may publish an updated limit with additional exposure.

A2: Section 6, Hyper-K timeline. The paper states operations begin “approximately 2027.” This is based on the collaboration’s publicly stated schedule. Large underground physics experiments frequently experience delays of 1-3 years. The paper’s use of “approximately” and “scheduled” is appropriate. A future session should verify the timeline if writing after 2027.

A3: Section 7, the “non-observation” path. The paper correctly states that non-observation excludes minimal SU(5) completion but not the Cabibbo Doublet itself. For completeness: threshold corrections at M_GUT are the most likely rescue mechanism. If the heavy X and Y bosons are split in mass (some heavier, some lighter than the nominal M_GUT), the effective M_GUT for proton decay can shift upward by a factor of 2-3, pushing the predicted lifetime from 10^35 into the 10^36-37 range — beyond Hyper-K. This is not fine-tuning; mass splittings of order 2-3 are generic in GUT models. The gap ratio constraint (38/27) applies to the AVERAGE unification scale, while the proton decay rate depends on the SPECIFIC mass of the X and Y bosons that mediate the decay. These need not be identical.

Annotation text for A3: “The threshold correction rescue mechanism works as follows: the gap ratio 38/27 constrains the scale where the three couplings approximately converge, but the proton decay rate depends on the specific masses of the X and Y gauge bosons. In a realistic GUT, these masses are split around the nominal M_GUT by factors of 2-3 due to GUT-scale symmetry breaking. If the X and Y masses are 2-3 $\times$ above the nominal M_GUT, the proton lifetime shifts by $(2\text{-}3)^4 = 16\text{-}81\times$, potentially from 10^35 to 10^36-37 yr — beyond Hyper-K but consistent with the gap ratio. This is the most conservative escape route for the Cabibbo Doublet scenario if Hyper-K sees nothing.”

Appendix A: The Chain from Integers to Experiment

A.1: Each Step with Source

Step	Input	Operation	Output	Source	Level
1	SM particle content	Beta coefficients b_1=41/10, b_2=−19/6, b_3=−7	SM gap ratio 218/115 = 1.896	GUT script, PASS	Level 1

Step	Input	Operation	Output	Source	Level
2	Cabibbo Doublet (3,2,1/6)	Δb_1=1/15, Δb_2=1, Δb_3=1/3	Modified gap ratio 38/27 = 1.407	GUT script, PASS	Level 1
3	Modified betas + DATA-3 couplings	One-loop running: $1/\alpha_1(\mu) = 1/\alpha_2(\mu)$ at μ = M GUT	M GUT = 10^15.5 GeV	GUT script, PASS	Level 1 + Level 2
4	M GUT in minimal SU(5)	$\tau \propto$ M GUT^4 / (α GUT^2 $\times$ m p^5 $\times$ matrix element^2)	τ ~ 10^34-35 yr	Standard GUT formula + lattice QCD	Level 1 + Level 2
5	τ ~ 10^34-35 yr	Compare to Super-K bound and Hyper-K sensitivity	Within Hyper-K reach	Web-verified experimental data	Level 2

Steps 1-2 are pure Level 1 — exact rational arithmetic on gauge group integers. Step 3 introduces Level 2 (the DATA-3 couplings). Step 4 introduces further Level 2 (hadronic matrix elements, α_GUT). Step 5 is entirely Level 2 (experimental parameters). The chain from integers to experiment crosses the Level 1 / Level 2 boundary at Step 3.

Appendix B: M_GUT Comparison — All Scenarios

B.1: From the Verified GUT Script

Scenario	Beta Coefficients	Gap Ratio	Exact Fraction	M GUT (GeV)	$\log_{10}$	Script Check
SM	(41/10, −19/6, −7)	1.896	218/115	6.3 × 10^13	13.80	PASS
SM + Cabibbo Doublet	(41/10+1/15, −19/6+1, −7+1/3)	1.407	38/27	3.2 × 10^15	15.50	PASS
SM + SU(5) 5+5̄	—	1.481	40/27	7.9 × 10^14	14.90	Excluded (proton decay)

Scenario	Beta Coefficients	Gap Ratio	Exact Fraction	M GUT (GeV)	$\log_{10}$	Script Check
Full MSSM	(33/5, 1, −3)	1.400	7/5	2.1 × 10^17	17.32	PASS

B.2: The Separation

Comparison	M GUT Ratio	M GUT^4 Ratio	Lifetime Ratio
Cabibbo Doublet vs SM	10^15.5 / 10^13.8 = 10^1.7 ≈ 50	50^4 ≈ 6.3 × 10^6	CD lifetime ~10^7 × SM lifetime
Cabibbo Doublet vs MSSM	10^17.3 / 10^15.5 = 10^1.8 ≈ 63	63^4 ≈ 1.6 × 10^7	MSSM lifetime ~10^7 × CD lifetime
Cabibbo Doublet vs SU(5) $5+\bar{5}$	10^15.5 / 10^14.9 = 10^0.6 ≈ 4	4^4 = 256	CD lifetime ~256 × $5+\bar{5}$ lifetime

The Cabibbo Doublet sits at the geometric midpoint (on a log scale) between the excluded SM and the untestable MSSM. This is the testability sweet spot.

Appendix C: The M_GUT^4 Scaling

C.1: Sensitivity Table

M GUT (GeV)	$\log_{10}$(M GUT)	τ (approximate)	Status
10^13.8	13.8	~10^30 yr	EXCLUDED (Super-K by 4 orders of magnitude)
10^14.5	14.5	~10^32 yr	EXCLUDED
10^14.9	14.9	~10^33 yr	EXCLUDED (SU(5) $5+\bar{5}$)
10^15.0	15.0	~10^33-34 yr	EXCLUDED
10^15.3	15.3	~3 × 10^33 yr	EXCLUDED
10^15.5	**15.5**	**~10^34-35 yr**	**AT BOUNDARY — Hyper-K window**
10^16.0	16.0	~10^36 yr	Beyond Hyper-K
10^16.5	16.5	~10^37 yr	Far beyond

M GUT (GeV)	$\log_{10}$(M GUT)	τ (approximate)	Status
10^17.3	17.3	~10^37-38 yr	MSSM — untestable by planned experiments

C.2: The Fourth Power Amplification

Change in M GUT	Factor in M GUT	Factor in τ (= M GUT^4)	Physical Meaning
+0.1 in $\log_{10}$	1.26 ×	$1.26^4 = 2.5\times$	Small running correction → factor 2.5 in lifetime
+0.3 in $\log_{10}$	2.0 ×	$2^4 = 16\times$	Cabibbo Doublet mass uncertainty → ~1 order of magnitude
+0.5 in $\log_{10}$	3.2 ×	$3.2^4 \approx 100\times$	Moderate threshold correction → two orders of magnitude
+1.0 in $\log_{10}$	10 ×	$10^4 = 10{,}000\times$	Major parameter change → four orders of magnitude
+1.8 in $\log_{10}$	63 ×	$63^4 \approx 1.6 \times$ 10^7	Cabibbo Doublet → MSSM: seven orders of magnitude

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Appendix D: The Proton Decay Process

D.1: $p \to e^+ \pi^0$ in Minimal SU(5)

Property	Detail
Initial state	Proton (uud), mass 938.272 MeV
Final state	Positron (e^+) + neutral pion (π^0)
Mediating particle	X or Y gauge boson (mass ~ M GUT)

Property	Detail
Mechanism	Two quarks exchange an X boson; one quark → positron, other quark → antiquark → pion
Operator dimension	Dimension 6 ($qq \rightarrow e^+\bar{q}$, suppressed by $1/M\ GUT^2$)
B−L conservation	B changes by −1, L changes by +1; B−L conserved
Electric charge	+1 → +1 + 0 (conserved)
π^0 decay	$\pi^0 \rightarrow \gamma\gamma$ (99% BR, $\tau = 8.5 \times 10^{-17}$ s)
Total visible energy	938 MeV (proton rest mass, no missing energy)

D.2: Experimental Signature in Water Cherenkov

Feature	What Is Detected
Positron	One electromagnetic Cherenkov ring
Photon 1 (from π^0)	One electromagnetic Cherenkov shower/ring
Photon 2 (from π^0)	One electromagnetic Cherenkov shower/ring
Total	Three rings, back-to-back topology
Invariant mass	Reconstructed to 938 MeV
Total momentum	Consistent with zero (proton at rest in water)
Background	Atmospheric neutrino CC single-pion production (very low rate in this topology)
Detection efficiency	~45% (Super-K estimate)

D.3: Why This Channel Is Clean

The three-ring topology at zero total momentum and proton invariant mass is almost unique. Atmospheric neutrinos can mimic it through charged-current single-pion production (ν _e + n → e^- + π^0 + p), but only if the proton is below the Cherenkov threshold (invisible) and the electron is misidentified as a positron. This background rate is approximately 0.5 events per megaton-year of exposure — extremely low. Super-K found zero candidates for $p \rightarrow e^+ \pi^0$ in 450 kton·years.

Appendix E: Super-Kamiokande — Current Status

E.1: Detector Specifications

Property	Value
Location	Mozumi mine, Gifu Prefecture, Japan
Depth	~1000 m rock overburden (2700 m.w.e.)
Total water	50 kton
Fiducial volume (original)	22.5 kton
Fiducial volume (enlarged, 2020 analysis)	27.2 kton
Photosensors	11,129 PMTs (20-inch, inner detector)
Photo-coverage	~40%
Operating since	April 1996
Total exposure (2020 analysis)	450 kton·years

E.2: Proton Decay Bounds ($p \rightarrow e^{+} \pi^{0}$)

Publication	Data Period	Exposure	Candidates	Bound (90% CL)
Phys. Rev. Lett. 102, 141801 (2009)	1996-2008	~220 kton·yr	0	τ > 1.3 × 10^34 yr
Phys. Rev. D 95, 012004 (2017)	1996-2015	306 kton·yr	0	τ > 1.6 × 10^34 yr
Phys. Rev. D 102, 112011 (2020)	**1996-2018**	**450 kton·yr**	**0**	**τ > 2.4 × 10^34 yr**

The bound has improved by approximately a factor of 2 over the past decade, driven by increased exposure time and enlarged fiducial volume. Zero candidates in all searches.

Appendix F: Hyper-Kamiokande — Specifications and Projections

F.1: Detector Comparison

Property	Super-K	Hyper-K	Ratio
Total water volume	50 kton	258 kton	5.2 ×
Fiducial volume	22.5 kton (original)	188 kton	8.3 ×
PMT type	R3600 (20-inch)	R12860 (20-inch, improved)	2 × quantum efficiency
Number of PMTs	11,129 (ID)	~20,000 (ID)	1.8 ×
Photo-coverage	~40%	~20% (but higher QE compensates)	Comparable effective coverage

Property	Super-K	Hyper-K	Ratio
Rock overburden	~1000 m	~650 m (1750 m.w.e.)	Slightly less
Location	Kamioka mine	Tochibora mine (8 km away)	Same region
Operations start	1996	~2027	31 years later

F.2: Projected Proton Decay Sensitivity ($p \rightarrow e^{+} \pi^{0}$)

Duration	Projected Limit (no observation)	Comparison to Super-K Current	Status vs Cabibbo Doublet
0 years (start)	$2.4 \times$ 10^34 yr (Super-K legacy)	Baseline	Lower end excluded
5 years	~4 $\times$ 10^34 yr	1.7 $\times$ improvement	Probing viable range
10 years	~6.3 $\times$ 10^34 yr	2.6 $\times$ improvement	Well into viable range
20 years	~10^35 yr	4 $\times$ improvement	**Full Cabibbo Doublet range covered**

F.3: Construction Timeline (Web-Verified)

Date	Milestone
January 2020	Budget approval, project officially begins
2020-2022	Access tunnel excavation (2 km)
June 2022	Access tunnel completed
2022-present	Cavern excavation and detector facility construction
~2027	Operations begin (first data for physics)
~2028	Full configuration with near detectors and beam
~2037	10-year exposure milestone
~2047	20-year exposure — full proton decay sensitivity

Appendix G: The Discriminator Matrix

G.1: Experimental Outcomes vs Scenarios

Observation	Cabibbo Doublet (min. SU(5))	Cabibbo Doublet (non-min. GUT)	MSSM	No Unification
$p \to e^+ \pi^0$ at τ ~ 10^34-35	**CONFIRMED**	Possible (depends on completion)	**EXCLUDED**	**EXCLUDED**
$p \to e^+ \pi^0$ at τ ~ 10^36-37	Excluded	Possible	**CONFIRMED**	Excluded
$p \to K^+ \bar{\nu}$ at τ ~ 10^34	Depends on completion	Possible (SO(10))	**CONFIRMED**	Excluded
No decay at τ > 10^35	**EXCLUDED**	Possible (threshold corrections)	Consistent	Consistent
VL quark found at LHC (1.5-3 TeV)	Strongly supports	Strongly supports	Not addressed	Not addressed
CKM deficit confirmed at 5σ	Supports	Supports	Not addressed	Not addressed

G.2: The Key Discriminations

Question	Answered By	When
Cabibbo Doublet vs MSSM?	Proton decay lifetime (τ differs by 10^7)	Hyper-K 2027-2037
Minimal SU(5) vs non-minimal GUT?	Proton decay channel ($e^+ \pi^0$ vs $K^+ \bar{\nu}$ vs others)	Hyper-K + DUNE 2027-2037
Cabibbo Doublet exists?	Direct production at LHC	HL-LHC now-2040
CKM deficit is real?	Improved V ud, V us precision	Belle II now-2030+

Appendix H: Model Dependence — Detailed

H.1: Sources of Uncertainty in the Proton Lifetime

Source	Effect on $\log_{10}(\tau)$	Effect on τ	Nature of Uncertainty
GUT completion group	Up to $\pm$ 2	Up to 100 $\times$	Structural: different groups give different operators
GUT threshold corrections	$\pm$ 0.5 to $\pm$ 1.0	3 $\times$ to 10 $\times$	Parametric: heavy particle mass splitting around M GUT
Hadronic matrix element	$\pm$ 0.3	~2 $\times$	Computational: lattice QCD uncertainty
Two-loop running	$\pm$ 0.1 to $\pm$ 0.2	1.3 $\times$ to 1.6 $\times$	Perturbative: known correction, small
α GUT uncertainty	$\pm$ 0.1	~1.3 $\times$	Parametric: from coupling running uncertainty
Cabibbo Doublet mass (1.5-6 TeV)	$\pm$ 0.5	~3 $\times$	Parametric: threshold correction from M VL
Combined (minimal SU(5))	**~34 to 35**	**~10^34 to 10^35**	**Order of magnitude range**

H.2: What Is Robust and What Is Not

Statement	Robust?	Why
M GUT = 10^15.5 from the gap ratio	Yes (at one loop)	Depends only on beta coefficients and DATA-3 couplings
Gap ratio = 38/27	Yes (exactly)	Pure Fraction arithmetic on integers
τ ~ 10^34-35 in minimal SU(5)	Moderately	Depends on GUT completion + matrix elements
"Within Hyper-K reach"	Yes	Robust across entire prediction range
$p \to e^+ \pi^0$ is dominant channel	Conditional	Only in minimal SU(5); changes in other completions
MSSM produces τ ~ 10^37	Yes	Well-established calculation in the literature

Statement	Robust?	Why
Cabibbo Doublet and MSSM are distinguishable	Yes	10^7 separation in lifetime, far exceeding uncertainties

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Appendix I: Complementary Experiments

I.1: Proton Decay Experiments

Experiment	Detector	Mass	Best Channel	Sensitivity	Timeline	Primary Mission
Hyper-K	Water Cherenkov	188 kton FV	**$p \to e^+ \pi^0$**	**~10^35 yr (20 yr)**	**2027+**	Neutrino CP violation
DUNE	Liquid Argon TPC	40 kton FV	**$p \to K^+ \bar{\nu}$**	~10^34 yr	2028+	Neutrino oscillations
JUNO	Liquid Scintillator	20 kton	$p \to K^+ \bar{\nu}$	~10^34 yr	2025+	Neutrino mass ordering

I.2: Channel-to-Scenario Mapping

Proton Decay Channel	Operator Dimension	GUT Completion	Primary Detector
$p \to e^+ \pi^0$	Dim-6 (gauge boson exchange)	Minimal SU(5), non-SUSY GUTs	Hyper-K
$p \to K^+ \bar{\nu}$	Dim-5 (colored Higgsino exchange)	SUSY SU(5), SUSY SO(10)	DUNE
$p \to \mu^+ \pi^0$	Dim-6	Some flipped SU(5) models	Hyper-K
$p \to e^+ K^0$	Dim-6	Some SO(10) models	Hyper-K / DUNE

If the Cabibbo Doublet scenario is correct AND the GUT completion is minimal SU(5): Hyper-K is the experiment. If the completion involves SUSY-like operators: DUNE becomes the primary probe. Together, the two experiments cover the dominant channels of both frameworks.

Appendix J: Timeline

J.1: Past and Future

Year	Event	Relevance
1974	Georgi & Glashow propose SU(5) GUT	Proton decay predicted
1983	IMB and Kamiokande begin proton decay searches	First experimental limits
1996	Super-Kamiokande begins operations	World-leading sensitivity
2000	LEP decommissioned; A FB^b frozen at 0.0992	Second Cabibbo Doublet anomaly established
2012	Higgs boson discovered at LHC	Third anomaly (μ excess) begins
2017	Super-K: τ > 1.6 × 10^34 yr	Minimal SU(5) without new physics excluded
2019-2020	CKM deficit identified as BSM signal	Anomaly path opens
2020	Super-K: **τ > 2.4 × 10^34 yr**	**Current bound**
2020	Hyper-K construction begins	Clock starts
2026	PHYS-15: Gap ratio identifies (3,2,1/6)	Two roads converge
~2027	**Hyper-K begins operations**	**Proton decay test begins**
~2028-2029	DUNE begins operations	Complementary channel
~2032	Hyper-K 5-year: τ > ~4 × 10^34 yr	Probing lower Cabibbo Doublet range
~2037	**Hyper-K 10-year: τ > ~6.3 × 10^34 yr**	**Well into Cabibbo Doublet range**
~2047	Hyper-K 20-year: τ > ~10^35 yr	Full Cabibbo Doublet range covered

J.2: The Decision Window

The critical decade is 2027-2037. Within this period:

Hyper-K begins collecting data (2027) and reaches sensitivity well into the Cabibbo Doublet prediction range (6.3 × 10^34 yr by 2037). If the proton lifetime is in the lower half of the viable range (~3 × 10^34 to ~5 × 10^34), observation could come within the first 5-10 years. If it is in the upper half (~5 × 10^34 to 10^35), the full 20-year dataset may be needed.

Simultaneously, the HL-LHC (operating through ~2040) searches for the Cabibbo Doublet directly in VL quark pair production, probing masses up to 2-3 TeV. Belle II sharpens the CKM deficit measurement. If all three programs produce positive results — proton decay at Hyper-K, VL quark at LHC, CKM deficit at Belle II — the Cabibbo Doublet is established from three independent experimental programs.

Appendix K: Level 1 / Level 2 Classification

K.1: What Is Level 1

Quantity	Value	Determination
SM beta coefficients	(41/10, −19/6, −7)	Gauge group representation theory
Cabibbo Doublet beta shifts	(1/15, 1, 1/3)	Dynkin index formulas for (3,2,1/6)
SM gap ratio	218/115	Exact Fraction arithmetic
Modified gap ratio	38/27	Exact Fraction arithmetic
$\tau \propto$ M GUT^4 scaling	Dimensional analysis	GUT operator structure (dim-6)
$p \to e^+ \pi^0$ dominance in minimal SU(5)	Operator classification	SU(5) representation theory

K.2: What Is Level 2

Quantity	Value	Source
α^{-1}, $\sin^2\theta$ W, α s at M Z	137.036, 0.23122, 0.1180	DATA-3
M GUT = 10^15.5	From running + DATA-3	Level 1 betas + Level 2 couplings
τ ~ 10^34-35 yr	From M GUT + formula	Level 1 scaling + Level 2 inputs
α GUT	~1/37 (from running)	Level 2
Hadronic matrix element	Lattice QCD	Level 2
Super-K bound 2.4 × 10^34	Experimental measurement	Level 2
Hyper-K sensitivity ~10^35	Projected from detector specifications	Level 2
Cabibbo Doublet mass 1.5-6 TeV	From anomaly fits	Level 2 (PHYS-19)

K.3: The Boundary

The prediction chain crosses the Level 1 / Level 2 boundary at the point where DATA-3 couplings enter the running equation. Everything before that point (the gap ratio 38/27, the beta coefficients, the scaling law) is Level 1. Everything after (M_GUT, the proton lifetime, the experimental comparison) involves Level 2 inputs.

The Level 1 content — the integers, the gap ratio, the scaling law — is permanent. It does not change with improved measurements. The Level 2 content — M_GUT, τ, the experimental bounds — will sharpen as measurements improve. The chain is designed so that Level 1 provides the structure and Level 2 fills in the numbers.

Appendix L: Verified Script Output

From sin2_theta_w_1.py (GUT running notebook), 9/9 checks:

```
[PASS] Normalization: sin²θ_W from couplings

   diff = 0.00e+00

[PASS] SM gap ratio = 218/115

   1.8956521739

[PASS] MSSM gap ratio = 7/5

   1.4000000000

[PASS] SM does not unify (Δ > 5)

   Δ(1/α₃) = -6.58

[PASS] MSSM nearly unifies (Δ < 5)

   Δ(1/α₃) = -0.69

[PASS] M_GUT(SM) > 10^13

   log₁₀ = 13.80

[PASS] M_GUT(MSSM) > 10^16

   log₁₀ = 17.32

[PASS] VL quark doublet gap < 0.05 from measured
```

```
   distance = 0.049

[PASS] Measured gap ratio in [1.2, 1.5]

   gap = 1.358193
```

From the enumeration table (Result 6):

Rank 1: Full MSSM — Gap = 1.4000, Dist = 0.0418, log M_GUT = 17.3. SAFE.

Rank 2: VL fermion (3,2,1/6) — Gap = 1.4074, Dist = 0.0492, log M_GUT = 15.5. SAFE.

Rank 3: SU(5) $5+\bar{5}$ fermion — Gap = 1.4815, Dist = 0.1233, log M_GUT = 14.9. EXCLUDED (proton decay).

Experimental parameters verified by web search April 2026:

Super-K: $\tau(p \to e^+ \pi^0)$ > 2.4 × 10^34 yr at 90% CL (Phys. Rev. D 102, 112011, 2020, arXiv:2010.16098). 450 kton·years exposure, 0 candidates.

Hyper-K: 258 kton total, 188 kton fiducial. Operations ~2027. Sensitivity ~10^35 yr for $p \to e^+ \pi^0$ with 20 years of data. 10-year projected limit 6.3 × 10^34 yr. (SciPost Phys. Proc. 17, 019, 2025; Hyper-K Design Report arXiv:1805.04163).

All measured values from DATA-3 (123 entries, 32/32 consistency checks pass).

Supporting appendix tables A through L for PHYS-20. Every M_GUT value traces to the verified GUT script (9/9 pass). Every experimental parameter verified by web search. The M_GUT^4 scaling table provides the quantitative framework for interpreting any future proton decay result. The discriminator matrix shows that the Cabibbo Doublet and the MSSM, despite nearly identical gap ratios, are separated by seven orders of magnitude in proton lifetime — a difference that Hyper-Kamiokande can resolve within one decade of operations.

The paper already contains Appendices A through L with comprehensive tables. My summary covered the errata and key content. The supporting appendix tables need to be NEW content providing deep reference material beyond what exists.

APPENDIX M: THE COMPLETE CHAIN — FROM THE INTEGER 11 TO HYPER-KAMIOKANDE

Every step from the Yang-Mills coefficient to the experimental test, with the type of each link and what would break it.

Step	Input	Operation	Output	Type	What Would Break This Link
1	Yang-Mills theory in 4D	One-loop computation	Gauge self-coupling coefficient $-11C_2(G)/3$	Mathematical theorem	Nothing — follows from Lorentz + gauge invariance
2	SU(3) × SU(2) × U(1) gauge group	Apply Step 1	Gauge betas: (0, −22/3, −11)	Theorem	Discovery that gauge group is different
3	SM fermion content (3 × 5+10 of SU(5))	Anomaly cancellation	Per-gen: (4/3, 4/3, 4/3) — democratic, invisible to gap ratio	Theorem	Discovery that SM fermions don't fill complete SU(5) reps
4	Higgs (1,2,1/2)	Scalar beta formula	(1/10, 1/6, 0) — democracy-breaking	Theorem	Discovery of different Higgs sector
5	Steps 2+3+4	Addition	SM betas: (41/10, −19/6, −7)	Exact arithmetic	Error in Steps 2-4
6	Step 5	Ratio of differences	SM gap ratio: 218/115 = 1.896	Exact arithmetic	Error in Step 5
7	α em, $\sin^2\theta$ W, α s (DATA-3)	GUT normalization	Measured gap ratio: 1.358	Measurement	Improved measurements change 1.358
8	Steps 6 vs 7	Comparison	40% miss — SM does not unify	Arithmetic	If improved measurements happen to give 1.896
9	15 representations within scope	Dynkin index formulas	15 sets of (Δb_1, Δb_2, Δb_3)	Theorem	Larger scope reveals new candidates

Step	Input	Operation	Output	Type	What Would Break This Link
10	SM betas + each Δb	Fraction arithmetic	15 modified gap ratios	Exact arithmetic	Error in Step 9
11	Modified gaps vs 1.358	Distance threshold 0.15	3 survive Stage 1	Criterion (stated)	Different threshold (stable under 0.05-0.20)
12	M GUT for survivors	Running equation + DATA-3	3 unification scales	Exact arithmetic + measurement	Improved α s changes M GUT
13	Super-K bound $\tau >$ 2.4×10^{34}	Proton decay exclusion	SU(5) $5+\bar{5}$ eliminated	Experimental data	Updated Super-K result
14	Step 13	Two survivors	MSSM (7/5) and Cabibbo Doublet (38/27)	Result	—
15	Cabibbo Doublet M GUT = $10^{15.5}$	$\tau \propto$ M GUT^4 in minimal SU(5)	$\tau \sim 10^{34\text{–}35}$ yr	GUT formula + lattice QCD	Different GUT completion
16	Hyper-K specifications	Projected sensitivity	~10^{35} yr (20 yr exposure)	Engineering projection	Construction delays or technical issues
17	**Steps 15 vs 16**	**Comparison**	**Prediction within experimental reach**	**The testable prediction**	**Universe decides**

Steps 1-6 are Level 1 (pure mathematics). Steps 7 and 12-16 involve Level 2 (measurements and experimental parameters). The chain crosses the Level 1 / Level 2 boundary at Step 7. Everything before Step 7 is permanent — it does not change with improved measurements. Everything after Step 7 will sharpen as measurements improve.

APPENDIX N: THE M_GUT^4 AMPLIFICATION — WHY SMALL GAP RATIO DIFFERENCES BECOME HUGE LIFETIME DIFFERENCES

The gap ratios of the Cabibbo Doublet and the MSSM differ by 0.007. Their proton lifetimes differ by 10^7. This appendix traces the amplification through two stages.

N.1: Stage 1 — Gap Ratio to M_GUT (Logarithmic Amplification)

M_GUT is determined by the scale where $1/\alpha_1 = 1/\alpha_2$. The running equation:

$\ln(M\_GUT/M\_Z) = 2\pi \times (1/\alpha_1 - 1/\alpha_2) / (b_1 - b_2)$

The measured quantity $(1/\alpha_1 - 1/\alpha_2) = 31.525$ is the same for all scenarios. Only $(b_1 - b_2)$ — the gap ratio numerator — changes.

Scenario	$b_1 - b_2$	ln(M GUT/M Z)	$\log_{10}$(M GUT)	M GUT (GeV)
SM	109/15 = 7.267	$2\pi \times$ 31.525/7.267 = 27.26	13.80	6.3×10^{13}
Cabibbo Doublet	19/3 = 6.333	$2\pi \times$ 31.525/6.333 = 31.27	15.54	3.5×10^{15}
MSSM	28/5 = 5.600	$2\pi \times$ 31.525/5.600 = 35.37	17.32	2.1×10^{17}

The gap ratio numerator changes from 6.333 (CD) to 5.600 (MSSM) — a 12% decrease. But this appears in a denominator that is divided into a fixed numerator (31.525), so a 12% decrease in the denominator produces a 13% increase in the logarithm: from 31.27 to 35.37. This translates to 1.8 decades in $\log_{10}$(M_GUT): from 15.5 to 17.3.

N.2: Stage 2 — M_GUT to Proton Lifetime (Fourth-Power Amplification)

$\tau \propto M\_GUT^4$. The 1.8 decades in $\log_{10}$(M_GUT) become 7.2 decades in $\log_{10}(\tau)$:

Step	Quantity	CD Value	MSSM Value	Ratio MSSM/CD
Gap ratio numerator	$b_1 - b_2$	19/3 = 6.333	28/5 = 5.600	0.884 (12% less)
ln(M GUT/M Z)	$2\pi \times 31.525/(b_1 - b_2)$	31.27	35.37	1.131 (13% more)
$\log_{10}$(M GUT)	—	15.54	17.32	+1.78 decades
M GUT	—	3.5×10^{15}	2.1×10^{17}	60 ×

Step	Quantity	CD Value	MSSM Value	Ratio MSSM/CD
M GUT^4	—	1.5×10^{62}	1.9×10^{69}	1.3×10^7
τ (proton lifetime)	—	~$10^{34.5}$	~$10^{41.7}$…	~10^7

The amplification chain: 12% in gap ratio numerator → 13% in ln(M_GUT) → 1.8 decades in $\log_{10}$(M_GUT) → 7.2 decades in $\log_{10}(\tau)$. A barely distinguishable difference in the gap ratio becomes a seven-order-of-magnitude separation in proton lifetime.

This is why proton decay is the decisive test. The gap ratios are 1.407 and 1.400 — practically indistinguishable. The proton lifetimes are $10^{34.5}$ and $10^{41.5}$ — completely distinguishable by any detector with megaton-year exposure.

APPENDIX O: ALL PROTON DECAY BOUNDS — HISTORICAL AND PROJECTED

Experiment	Detector Type	Exposure	Channel	Bound τ (90% CL, years)	Year
IMB	Water Cherenkov, 8 kton	~100 kton·yr	$p \to e^+ \pi^0$	$> 5.5 \times 10^{32}$	1991
Kamiokande	Water Cherenkov, 3 kton	~50 kton·yr	$p \to e^+ \pi^0$	$> 6 \times 10^{32}$	1993
Super-K (Phase I)	Water Cherenkov, 22.5 kton	92 kton·yr	$p \to e^+ \pi^0$	$> 5.4 \times 10^{33}$	2005
Super-K (Phase I+II)	Same	220 kton·yr	$p \to e^+ \pi^0$	$> 1.3 \times 10^{34}$	2009
Super-K (Phase I-IV)	Same, enlarged FV	306 kton·yr	$p \to e^+ \pi^0$	$> 1.6 \times 10^{34}$	2017
Super-K (current)	**Same**	**450 kton·yr**	$\mathbf{p \to e^+ \pi^0}$	$\mathbf{> 2.4 \times 10^{34}}$	**2020**
Hyper-K (5 yr projected)	Water Cherenkov, 188 kton	~940 kton·yr	$p \to e^+ \pi^0$	> ~4×10^{34}	~2032
Hyper-K (10 yr projected)	Same	~1880 kton·yr	$p \to e^+ \pi^0$	> ~6.3×10^{34}	~2037

Experiment	Detector Type	Exposure	Channel	Bound τ (90% CL, years)	Year
Hyper-K (20 yr projected)	**Same**	**~3760 kton·yr**	**$p \to e^{+} \pi^{0}$**	**> ~10^{35}**	**~2047**
Super-K (current)	Same	450 kton·yr	$p \to K^{+} \bar{\nu}$	> 6.6×10^{33}	2014
DUNE (10 yr projected)	Liquid Argon, 40 kton	~400 kton·yr	$p \to K^{+} \bar{\nu}$	> ~10^{34}	~2038

The improvement has been steady: from 5×10^{32} (IMB, 1991) to 2.4×10^{34} (Super-K, 2020) — a factor of ~50 in three decades. Hyper-K adds another factor of ~4 with 20 years of running. The Cabibbo Doublet prediction sits exactly at the frontier where decades of improvement have brought us.

APPENDIX P: THE TESTABILITY LANDSCAPE — EVERY ENUMERATED CANDIDATE PLACED

Every candidate from the PHYS-15 enumeration, placed in the proton decay landscape.

Rank	Candidate	Gap Ratio	M GUT ($\log_{10}$)	Predicted τ	Super-K Status	Hyper-K Status	Testable?
1	Full MSSM	7/5 = 1.400	17.3	~10^{37} yr	Safe	Safe	No — beyond any planned experiment
2	**Cabibbo Doublet (3,2,1/6)**	**38/27 = 1.407**	**15.5**	**~$10^{34\text{-}35}$ yr**	**At boundary**	**Within reach**	**YES — Hyper-K 2027-2047**
3	SU(5) 5+5	40/27 = 1.481	14.9	~10^{32} yr	EXCLUDED	—	Already dead
4	3 × Scalar (1,2,1/2)	1.631	14.1	~10^{30} yr	EXCLUDED	—	Already dead
5	Scalar (3,2,1/6)	1.632	14.6	~10^{31} yr	EXCLUDED	—	Already dead

Rank	Candidate	Gap Ratio	M GUT ($\log_{10}$)	Predicted τ	Super-K Status	Hyper-K Status	Testable?
6	Scalar (1,3,0)	1.664	14.4	~10^{31} yr	EXCLUDED	—	Already dead
7	VL (1,2,−1/2)	1.712	14.0	~10^{29} yr	EXCLUDED	—	Already dead
8	2 × Scalar (1,2,1/2)	1.712	14.0	~10^{29} yr	EXCLUDED	—	Already dead
9	Scalar (1,2,1/2)	1.800	13.9	~10^{29} yr	EXCLUDED	—	Already dead
10	SU(5) 10+$\overline{10}$	1.948	13.5	~10^{27} yr	EXCLUDED	—	Already dead
11-15	Various	2.0-2.2	12.3-13.7	<10^{28} yr	EXCLUDED	—	Already dead

The landscape divides into three zones:

Zone 1 ($\tau < 10^{34}$): Already excluded by Super-K. Contains 13 of 15 candidates. Dead.

Zone 2 ($10^{34} < \tau < 10^{35}$): The Hyper-K window. Contains the Cabibbo Doublet. Testable.

Zone 3 ($\tau > 10^{35}$): Beyond any planned experiment. Contains the MSSM. Untestable for the foreseeable future.

Only one candidate sits in Zone 2. The Cabibbo Doublet is the uniquely testable survivor.

APPENDIX Q: THE SENSITIVITY SCALING — HOW DETECTOR SIZE MAPS TO PROTON LIFETIME REACH

Quantity	Formula	Value
Expected events	N = (N p × ε × t) / τ	N p = protons, ε = efficiency, t = time, τ = lifetime
Protons per kiloton water	6.0×10^{32}	From N A × (10^6 g) × (10/18) protons per water molecule
Super-K fiducial mass	22.5 kton (original)	1.35×10^{34} protons
Hyper-K fiducial mass	188 kton	1.13×10^{35} protons
Detection efficiency ($p \to e^+ \pi^0$)	~45%	Water Cherenkov estimate

Quantity	Formula	Value
Background rate (Super-K)	~0.5 events per Mton·yr	Atmospheric neutrino CC π^0

Q.1: Sensitivity vs Exposure Time

For a Poisson counting experiment with near-zero background, the 90% CL limit with zero observed events is 2.3 events:

τ_limit = N_p × ε × t / 2.3

Experiment	N p × ε	Time	τ limit	Matches Published?
Super-K (450 kton·yr)	22.5 kton × 0.45 = 10.1 kton effective	20 yr	$10.1 \times 6 \times 10^{32} \times 20 / 2.3 = 5.3 \times 10^{34}$	Published 2.4×10^{34} — factor 2 from refined analysis cuts
Hyper-K (10 yr)	188 × 0.45 = 84.6 kton effective	10 yr	$84.6 \times 6 \times 10^{32} \times 10 / 2.3 = 2.2 \times 10^{35}$	Projected ~6.3×10^{34} — factor 3 from backgrounds + analysis
Hyper-K (20 yr)	Same	20 yr	4.4×10^{35}	Projected ~10^{35} — factor 4 from backgrounds

The simple formula overestimates sensitivity by factors of 2-4 because it neglects background subtraction, energy resolution, and systematic uncertainties. The collaboration projections account for these. The scaling with mass × time is correct; the absolute normalization requires detailed simulation.

Q.2: What Observation Would Look Like

Scenario	τ (years)	Events in Hyper-K (10 yr)	Events in Hyper-K (20 yr)	Significance
$\tau = 3 \times 10^{34}$	Low end of CD prediction	~5-8 events	~10-16 events	Discovery (>5σ)
$\tau = 10^{35}$	High end of CD prediction	~1-2 events	~2-4 events	Evidence (2-3σ)

Scenario	τ (years)	Events in Hyper-K (10 yr)	Events in Hyper-K (20 yr)	Significance
$\tau = 3 \times 10^{35}$	Above CD prediction	~0.3-0.5 events	~0.7-1 events	Hint or null
$\tau = 10^{37}$	MSSM prediction	~0.01 events	~0.02 events	Null (undetectable)

If the proton lifetime is in the lower half of the Cabibbo Doublet range (3×10^{34} yr), Hyper-K discovers proton decay with 5σ significance within 10 years. If it is in the upper half (~10^{35} yr), the full 20-year dataset is needed and the significance may be only 2-3σ — evidence rather than discovery.

APPENDIX R: THE COMPLETE SERIES — WHAT EACH PAPER CONTRIBUTES TO THIS PREDICTION

Paper	Key Result	Role in Proton Decay Prediction
MATH-1	$R_2 = \pi/4$ in nine engineering domains	π in beta coefficients is $4R_2$
MATH-2	17 transcendentals as exact integer pairs	Foundation for Fraction arithmetic
MATH-3	Extended basis: elliptic integrals, Borwein $\zeta(5)$	Seeds 4-loop wall resolution
MATH-4	Universal Q335 denominator	Computational infrastructure
MATH-5	$R_4 = \pi^2/32$, n-ball remainder	π^2 in loop factors is $32R_4$
MATH-6	82/82 PSLQ independence	SM constants don't decompose into basis — gap ratio miss is real
PHYS-1	Mass = inertia, soliton boundaries	Each mass threshold is a boundary
PHYS-2	Transformation laws are integers	Beta coefficients ARE the integers
PHYS-3	G untested outside Hill sphere	Not directly used
PHYS-4	Test program, kill switch	Not directly used
PHYS-5	α running at 0.02 ppm	Infrastructure for coupling running
PHYS-6	Confinement two-face	Hadronic VP limits gap ratio precision

Paper	Key Result	Role in Proton Decay Prediction
PHYS-7	θ QCD = 0 (19→18)	Parameter count baseline
PHYS-8	Koide (18→17 conditional)	Parameter count
PHYS-9	α from a e, 4.3 ppb	Demonstrates integer structure of QED
PHYS-10	Remainder framework, PSLQ null	SM couplings don't decompose — gap ratio miss is genuine
PHYS-11	Nine domains, R_2 universal	$2\pi = 8R_2$ in running equation
PHYS-12	Electroweak integer anatomy	Infrastructure: G F, M Z, $\sin^2\theta$ W as exact Fractions; R b overshoot = A FB^b physics
PHYS-13	Gap ratio 218/115, BSM enumeration	**The enumeration — finds (3,2,1/6)**
PHYS-14	Fermion cancellation theorem	**Why fermions don't matter — only bosons and BSM**
PHYS-15	Complete elimination cascade	**Two survivors: MSSM and Cabibbo Doublet**
PHYS-16	Cabibbo Doublet specification	**Named, quantum numbers, two roads**
PHYS-17	Generation democracy proved	**Why adding fermions can't fix unification**
PHYS-18	Y = 1/6 asymmetry mechanism	**Why the Cabibbo Doublet works — $\Delta b_2/\Delta b_1 = 15$**
PHYS-19	Three independent anomalies	**Bottom-up evidence: CKM, A FB^b, Higgs μ**
PHYS-20	**Proton decay test**	**$\tau \sim 10^{34\text{–}35}$ yr → Hyper-K 2027-2037**

Every paper contributes to the final prediction. The Fraction arithmetic infrastructure (MATH-2, MATH-4), the structural insights (PHYS-14 fermion cancellation, PHYS-17 generation democracy, PHYS-18 Y = 1/6 asymmetry), the precision computations (PHYS-5, PHYS-12), the enumeration (PHYS-13, 15), the anomaly evidence (PHYS-19), and the mechanism (PHYS-18) all flow into PHYS-20's single prediction: one experiment, one decade, one answer.

APPENDIX S: THE FINAL SCORECARD — EVERYTHING THE SERIES ESTABLISHES

Category	Result	Paper	Status	Conditional?
Parameter reductions				
θ QCD = 0	19 → 18	PHYS-7	Established	No — unconditional
m τ from Koide	18 → 17	PHYS-8	At 0.91σ from PDG	Yes — conditional on Koide exactness
α ↔ a e	Relabeling	PHYS-9	Demonstrated	N/A — not a reduction
Structural findings				
$R_2 = \pi/4$ universal	Present in 18 domains (9 engineering + 9 physics)	MATH-1, PHYS-11	Verified	No
139 PSLQ tests null	Transcendental basis is minimal	MATH-6, PHYS-10	Verified	No
Generation democracy	(4/3, 4/3, 4/3) per gen; fermions invisible to gap ratio	PHYS-14, PHYS-17	Proved algebraically	No
Gap ratio = boson problem	96-101% gauge, 0% fermion, −1% to +4% Higgs	PHYS-17	Proved	No
A_2 decomposition	87.4% cancellation between geometry and arithmetic	PHYS-12	Verified (9/9)	No
BSM identification				
Gap ratio 218/115 misses by 40%	SM does not unify	PHYS-13	Verified (9/9)	No

Category	Result	Paper	Status	Conditional?
15 candidates enumerated, 13 eliminated	Two survivors: MSSM and Cabibbo Doublet	PHYS-15	Verified (9/9)	Scope-dependent
Y = 1/6 produces maximum asymmetry $\Delta b_2/\Delta b_1 = 15$	Mechanism for gap ratio fix	PHYS-18	Proved	No
Three independent anomalies point to (3,2,1/6)	CKM 2.5-4σ, A FB^b ~3σ, Higgs ~2σ	PHYS-19	Documented	Each anomaly could have mundane explanation
Testable predictions				
$\tau(p \to e^+ \pi^0) \sim 10^{34\text{–}35}$ yr	Hyper-K 2027-2037	PHYS-20	Prediction	Conditional on minimal SU(5) completion
Mass 1.5-6 TeV	HL-LHC now-2040	PHYS-16, 19	Window	From anomaly constraints
CKM deficit	V ub'	≈ 0.045	Belle II now-2030+	Prediction

What the series has done: Built exact rational arithmetic infrastructure from transcendental constants through QED through electroweak physics to grand unification. Identified one particle — the Cabibbo Doublet — by two independent methods. Specified it completely. Connected it to three experimental anomalies. Derived a testable proton decay prediction. Documented every step in exact Fraction arithmetic with verified scripts.

What the series has NOT done: Derived any SM coupling constant from first principles. Explained why the gauge group is SU(3) × SU(2) × U(1). Derived the Cabibbo Doublet mass. Proved the Cabibbo Doublet exists. These remain open. The series identifies what the integers determine and where the universe must speak.

[**HOWL-PHYS-20-2026**] The Proton Decay Test — Hyper-Kamiokande and the Cabibbo Doublet at M_GUT = 10^15.5. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-20-2026

[**HOWL-PHYS-1-2026**] The Inertial Vortex. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-1-2026

[**HOWL-PHYS-2-2026**] There Are No Constants. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-2-2026

[**HOWL-PHYS-6-2026**] The Confinement Boundary in Integer Arithmetic. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-6-2026

[**HOWL-PHYS-7-2026**] The Strong CP Problem Is Not a Problem. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-7-2026

[**HOWL-PHYS-8-2026**] The Koide Constant Decomposes. Github: https://github.com/ghowland/papers/tree/main/papers/
PHYS-8-2026

[**HOWL-PHYS-9-2026**] The Electromagnetic Chain in Integer Arithmetic. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-9-2026

[**HOWL-PHYS-10-2026**] Remainder as Observable. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS
PHYS-10-2026

[**HOWL-PHYS-11-2026**] Remainder Structure Across Nine Physics Domains. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-11-2026

[**HOWL-PHYS-12-2026**] Electroweak Integer Anatomy. Github: https://github.com/ghowland/papers/tree/main/papers/P
PHYS-12-2026

[**HOWL-PHYS-13-2026**] Gauge Coupling Unification and Minimal BSM Content. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026

[**HOWL-PHYS-14-2026**] The Unified Transformation Map. Github: https://github.com/ghowland/papers/tree/main/paper
PHYS-14-2026

[**HOWL-PHYS-15-2026**] Integer-Forced Identification of the Minimal Unification Extension. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-15-2026

[**HOWL-PHYS-17-2026**] The Generation Democracy and the Boson Problem. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-17-2026

[**HOWL-PHYS-18-2026**] The Y = 1/6 Asymmetry. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/
PHYS-18-2026

[**HOWL-PHYS-19-2026**] Independent Anomaly Evidence for the Cabibbo Doublet. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-19-2026